Benchmarking Knowledge Retrieval Architectures Across 25 Domains: RAG, GraphRAG, and Compressed Knowledge Graphs on the McCreary Intelligent Textbook Corpus

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Abstract

Retrieval-augmented generation (RAG) and graph-based retrieval (GraphRAG) are the dominant paradigms for grounding LLM responses in structured knowledge. Both optimize for retrieval recall while treating token cost as a secondary concern. We introduce Compressed Knowledge Graphs (CKG)—pre-structured DAG representations with explicit concept taxonomy and pipe-delimited dependency encoding—and present the first large-scale benchmark comparing all three architectures across 25 domains.

We evaluate on the McCreary Intelligent Textbook Corpus: 25 open-source educational domains (calculus, genetics, bioinformatics, economics, quantum computing, and more), each with a standardized learning graph CSV totaling approximately 6,000 concepts and 8,500 dependency edges. We introduce five novel metrics: Reasoning Density Score (RDS = F1 / tokens), Context Utilization Rate, Hop-Depth F1 Degradation, Cost Per Correct Answer, and Relationship Precision.

CKG achieves macro-average F1 of X.XX vs X.XX (RAG) and X.XX (GraphRAG), at Xx fewer tokens, with largest gains on multi-hop dependency queries (T3) where RAG F1 degrades by XX% per additional hop while CKG remains flat. RDS ratio: CKG vs RAG = Xx, CKG vs GraphRAG = Xx.

Benchmark, dataset, and evaluation harness are released at `https://huggingface.co/datasets/graphify-md/ckg-benchmark` under CC BY 4.0 / MIT.

1 Introduction

1.1 Motivation

LLM retrieval quality is typically measured by F1 alone, yet token cost is a first-class production constraint—not a research afterthought. Domain-specific knowledge has latent structure that RAG discards through chunking, and GraphRAG re-derives structure from text at significant computational expense. But what if the structure is already known?

1.2 The Three Paradigms

Table 1 summarizes the three retrieval architectures compared in this benchmark.

*Graphify.md — `https://graphify.md`

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Table 1: Three knowledge retrieval paradigms compared in this benchmark.

System	Knowledge Repr.	Retrieval	Build Cost
RAG	Unstructured text chunks	Embedding similarity	Embed all chunks
GraphRAG	Dynamically extracted graph	Graph + community search	Full entity extraction
CKG	Pre-structured DAG + taxonomy	Direct concept/edge lookup	Zero (CSV-native)

1.3 Falsifiable Claims

We make the following falsifiable claims, each testable against the benchmark results presented in Section 7:

1. CKG achieves higher F1 on T2 (dependency) and T3 (multi-hop path) queries.
2. CKG F1 does not degrade with hop depth; RAG F1 degrades significantly.
3. CKG RDS ratio $\geq$ 10x vs RAG across all 25 domains.
4. GraphRAG hallucinates edges not present in ground truth DAG (HR > 0).
5. CKG Hallucination Rate $= 0$ (by construction).
6. The "Structure Premium" hypothesis: RDS advantage correlates with DAG richness ($r > 0.7$).

1.4 Contributions

This paper makes the following contributions:

1. The CKG architecture specification (format, DAG constraints, taxonomy schema).
2. Five novel evaluation metrics (RDS, CUR, Hop-F1, CPCA, RP).
3. The McCreary Corpus as a formal benchmark dataset (first paper to do so).
4. An open benchmark: 25 domains $\times$ $\sim$175 queries $\times$ 3 systems ($\sim$13,000 evaluated pairs).
5. A HuggingFace dataset with one-command reproduction harness.

2 Related Work

2.1 Retrieval-Augmented Generation

Lewis et al. [1] introduced RAG as a method for grounding language model outputs in retrieved passages. Dense Passage Retrieval (DPR) [2] established embedding-based retrieval as the dominant paradigm. The BEIR benchmark [3] provided heterogeneous evaluation across IR tasks, and RAGAS [4] introduced evaluation metrics for RAG pipelines including faithfulness and relevance.

2.2 Graph-Based Retrieval

Edge et al. [5] introduced GraphRAG, using LLM-extracted entity graphs with community detection for multi-hop reasoning. LightRAG [6] and HippoRAG proposed lighter-weight graph-based alternatives. The key distinction for this benchmark: these systems perform *dynamic extraction* from text, while CKG uses *pre-structured domain knowledge*.

2.3 Knowledge Graphs for LLMs

Pan et al. [7] provide a comprehensive roadmap for unifying LLMs and knowledge graphs. The KGQA literature demonstrates structured query answering over knowledge graphs such as Wikidata and UMLS. CKG differs from these approaches by using lightweight DAG representations rather than full ontologies, trading expressiveness for efficiency.

2.4 Evaluation Gaps in IR

Standard IR metrics (F1, MRR, NDCG) do not account for token cost. RAGAS measures faithfulness and relevance but not efficiency. This paper introduces Reasoning Density Score (RDS) and validates it on a multi-domain corpus—the first metric to jointly optimize quality and token consumption.

2.5 Educational Knowledge Graphs

McCreary [8] introduced the Intelligent Textbooks methodology, using learning graph DAGs as structured domain knowledge with a standardized CSV schema: `ConceptID | ConceptLabel | Dependencies | TaxonomyID`. This work is the first to formalize this corpus as a citable benchmark dataset.

3 The McCreary Intelligent Textbook Corpus

This section formally defines the corpus for the first time in literature.

3.1 Corpus Description

The McCreary Intelligent Textbook Corpus comprises 25 open-source educational textbooks hosted on GitHub (`https://github.com/dmccreary`). Each textbook contains a standardized learning graph encoded as a CSV file representing a directed acyclic graph (DAG) of concepts and their prerequisite dependencies.

The corpus spans three categories:

- **STEM** (11 domains): calculus, biology, genetics, bioinformatics, statistics, quantum computing, circuits, geometry, ecology, moss, functions.
- **Professional** (7 domains): economics, organizational analytics, healthcare data modeling, database selection, conversational AI, automating instructional design, blockchain.
- **Foundational** (7 domains): systems thinking, theory of knowledge, digital citizenship, prompt engineering, AI tracking, US geography, ASL.

Key statistics:

- Concepts per domain: 200–380 (mean: ∼240)
- Total concepts: ∼6,000
- Total edges: ∼8,500
- Taxonomy categories: 10–17 per domain
- Raw content: MkDocs Markdown chapters (∼3,000–8,000 words per textbook)

3.2 Corpus Schema

All 25 domains share an identical CSV schema:

```
ConceptID,ConceptLabel,Dependencies,TaxonomyID
1,Function,,FOUND
2,Domain and Range,1,FOUND
3,Function Notation,1,FOUND
4,Composite Function,1|3,FOUND
```

Listing 1: Learning graph CSV schema (calculus example).

Dependencies are pipe-delimited integer references to prerequisite `ConceptID` values. `TaxonomyID` assigns each concept to a domain-specific category (e.g., `FOUND` for foundational, `CORE` for core, `ADV` for advanced).

3.3 Quality Properties

All 25 DAGs are validated for the following structural properties:

1. Single connected component (no isolated subgraphs).
2. No self-references (no concept lists itself as a dependency).
3. Foundational concepts (zero prerequisites) ≥ 2 per domain.
4. Maximum dependency chain length reported per domain.

4 Architecture Specifications

All three systems use Claude Sonnet 4.6 at temperature=0 for generation, ensuring fair comparison. The systems differ only in how knowledge is stored and retrieved.

4.1 RAG Baseline

4.2 GraphRAG

4.3 CKG (Compressed Knowledge Graph)

Key distinction. GraphRAG re-derives structure from text that was originally generated *from* the learning graph CSV. CKG uses the graph directly. The efficiency gap is structural, not incidental.

Table 2: RAG baseline configuration.

Parameter	**Value**
Source	MkDocs `.md` chapters per textbook
Chunking	512 tokens, 50-token overlap
Embeddings	`text-embedding-3-small` (OpenAI)
Index	FAISS flat L2
Retrieval	Top-5 chunks
Generation	Claude Sonnet 4.6, temperature=0

Table 3: GraphRAG configuration.

Parameter	**Value**
Source	Same MkDocs `.md` chapters
System	Microsoft GraphRAG v1.x, default configuration
Search	Local mode for T1/T2/T5, global mode for T4
Note	Does **not** use `learning-graph.csv`
Generation	Claude Sonnet 4.6, temperature=0

Table 4: CKG architecture configuration.

Parameter	**Value**
Source	`learning-graph.csv`
Lookup	Exact label match → concept node retrieval
Traversal	BFS for T2 (1-hop), DFS for T3 (full path), filter for T4
Subgraph	Matched concept + direct neighbors + edges
Generation	Claude Sonnet 4.6, temperature=0
Note	Zero build cost—CSV-native

5 Benchmark Design

5.1 Query Taxonomy

We define five query types (T1–T5), each targeting a different aspect of knowledge retrieval capability. Table 5 summarizes the taxonomy.

Table 5: Query type taxonomy with examples and ground truth sources.

Type	Description	Example	Ground Truth
T1	Entity lookup	"What is Composite Function?"	ConceptLabel + TaxonomyID
T2	Direct dependency	"What are prerequisites for Composite Function?"	Dependencies column
T3	Multi-hop path	"What is the chain from Function to Taylor Series?"	BFS path in DAG
T4	Category aggregate	"List all FOUND concepts"	Filter by TaxonomyID
T5	Cross-concept	"How does Domain and Range relate to Inverse Function?"	Shared neighbors

5.2 Query Generation

Queries are auto-generated from each domain's CSV using `generate_queries.py` with a fixed random seed of 42.

Per domain: ∼175 queries (50 T1 + 50 T2 + 25 T3 + 12 T4 + 38 T5). Total: ∼4,375 queries across 25 domains.

- **T1**: Random sample of 50 concepts; query is "What is {label}?"
- **T2**: Random sample of 50 concepts with ≥1 dependency
- **T3**: Random pairs of foundational/terminal concepts with path length 2–5
- **T4**: One query per taxonomy category
- **T5**: Random sample of 38 directly connected concept pairs

5.3 Human Validation

A 5% sample (∼220 queries) is reviewed by two annotators. Cohen's κ is reported for inter-annotator agreement. DAG validation failures are handled and documented per domain.

5.4 Reproducibility Protocol

- All systems use Claude Sonnet 4.6 at temperature=0.
- Token counts via Anthropic `count_tokens()` API.

- 3 runs per query, variance reported.
- Fixed random seed: 42.
- Benchmark version locked: v1.0.0.
- One-command reproduction: `python evaluation/harness.py -reproduce-table-1`

6 Metrics

We evaluate 16 metrics organized into six categories. Novel metrics introduced in this paper are marked with $\star$.

6.1 Standard IR Metrics

Token-Level F1 (SQuAD-style). Used for T1, T2, and T4 queries. Precision, recall, and F1 are computed over token sets:

$$\text{F1} = \frac{2 \cdot P \cdot R}{P + R}, \quad P = \frac{|pred \cap truth|}{|pred|}, \quad R = \frac{|pred \cap truth|}{|truth|} \tag{1}$$

Edge-Overlap F1. Used for T3 (path) and T5 (cross-concept) queries:

$$\text{Edge_F1} = \frac{2 \cdot |E_{pred} \cap E_{truth}|}{|E_{pred}| + |E_{truth}|} \tag{2}$$

Exact Match (EM). Binary—the full answer must match ground truth exactly. Reported alongside F1 as a secondary metric.

6.2 Reasoning Density Score* (RDS)

The core compound metric introduced in this paper:

$$\text{RDS}(s, q) = \frac{\text{F1}(s, q)}{\text{tokens_consumed}(s, q)} \tag{3}$$

Macro-averaged across all queries: $\text{RDS}_{\text{macro}}(s) = \text{mean}(\text{RDS over all queries})$. The **RDS ratio** compares two systems: $\text{RDS}_{\text{ratio}}(A, B) = \text{RDS}_{\text{macro}}(A)/\text{RDS}_{\text{macro}}(B)$. Higher values indicate more reasoning quality per token spent.

6.3 Hop-Depth F1 Degradation*

Measures how F1 degrades as reasoning chain length increases:

$$\text{F1@hop}(s, k) = \text{mean}(\text{F1} \mid \text{hop_depth} = k) \tag{4}$$

Reported for $k = 1, 2, 3, 4, 5+$. Expected finding: RAG degrades steeply at $k \geq 2$; CKG remains flat due to explicit edge traversal.

6.4 Tokenomics Metrics

Context Utilization Rate* (CUR). Fraction of retrieved tokens relevant to the answer: CUR = relevant_tokens/total_retrieved_tokens.

Cost Per Correct Answer⋆ (CPCA). Real-world cost using Claude Sonnet 4.6 pricing ($3/M input, $15/M output): CPCA = cost_per_query/F1.

Precision at Token Budget (P@T). Mean F1 over queries where tokens_consumed $\leq$ budget T. Reported for $T = 500, 1000, 2000, 5000, 10000$.

Token Budget Breakeven. Minimum budget where RAG/GraphRAG F1 $\geq$ CKG F1: breakeven = $\min T$ such that $\mathrm{F1}_{\mathrm{RAG}}(T) \geq \mathrm{F1}_{\mathrm{CKG}}(500)$.

Index Build Cost. One-time cost: tokens consumed during indexing + wall-clock time + storage. CKG: zero (CSV already exists).

Update Cost⋆. Cost to incorporate one new concept: CKG edits one CSV row with zero re-indexing; RAG re-embeds affected chunks; GraphRAG requires full re-extraction.

6.5 Structural Fidelity Metrics

Relationship Precision⋆ (RP). Of edges returned, what fraction are real DAG edges: RP = $|E_{pred} \cap E_{truth}|/|E_{pred}|$.

Hub Node Recall⋆ (HNR). Recall on high-indegree concepts (top 20% by indegree).

Boundary Completeness⋆ (BC). For T4 queries, fraction of the taxonomy category returned: BC = $|retrieved \cap members|/|members|$. CKG achieves BC $\approx$ 1.0 by construction.

6.6 Robustness Metrics

Paraphrase Stability (PS). F1 variance across 5 paraphrased versions of each query. CKG should be stable (exact concept match); RAG is embedding-sensitive.

Hallucination Rate (HR). Fraction of queries returning $\geq$1 concept not in the corpus. CKG: HR = 0 by construction. GraphRAG's dynamic extraction can hallucinate.

7 Results

7.1 Macro-Average F1 Across 25 Domains

Table 6: Macro-average performance across 25 domains.

System	F1	EM	Tokens/q	RDS	CPCA ($)
RAG	—	—	—	—	—
GraphRAG	—	—	—	—	—
CKG	—	—	—	—	—

Table 7: F1 by query type (T1–T5).

System	T1	T2	T3	T4	T5
RAG	—	—	—	—	—
GraphRAG	—	—	—	—	—
CKG	—	—	—	—	—

7.2 F1 by Query Type

7.3 Token Cost and RDS

7.4 F1 by Hop Depth

7.5 Structural Metrics

7.6 Efficiency Curves

7.7 Structure Premium Correlation

8 Discussion

8.1 Where CKG Wins and Why

CKG's advantages are structural:

- **T2/T3 queries**: Explicit edges eliminate multi-hop inference errors. RAG must infer transitive dependencies from unstructured text; CKG traverses them directly via BFS/DFS.
- **T4 queries**: Taxonomy filtering achieves BC ≈ 1.0 by construction, as the TaxonomyID field provides exact category membership.
- **Hallucination**: HR $= 0$ because CKG only returns concepts present in the source CSV. No generative step can introduce phantom entities.
- **RDS**: Near-zero build cost combined with 150–400 tokens per query yields order-of-magnitude efficiency gains.

8.2 Where RAG Is Competitive

RAG remains competitive in specific scenarios:

- T1 entity lookup on large open-domain corpora where rich context aids natural language generation.
- Domains without stable taxonomy (rapidly evolving fields).
- When CKG construction cost exceeds the efficiency savings.

8.3 GraphRAG's Position

GraphRAG occupies a middle ground: better than RAG on multi-hop reasoning (graph structure helps) but worse than CKG (dynamic extraction introduces noise and hallucinated edges). GraphRAG is the most expensive system (high build cost + high query cost). Its best use case is unstructured corpora with no available expert taxonomy.

8.4 The Structure Premium

We hypothesize that the token efficiency gap between CKG and RAG is proportional to the structural richness of the domain's DAG:

$$\text{dag_richness}(d) = \frac{\text{edges}}{\text{concepts}} \times \text{mean_indegree} \times \frac{1}{\text{orphan_rate}} \tag{5}$$

If the correlation between structure_premium and dag_richness exceeds $r = 0.7$, this validates the core theoretical claim: explicit structure is the source of the efficiency gain, not domain selection bias.

8.5 Limitations

- The McCreary corpus is educational—results may not generalize to legal, financial, or medical domains.
- CKG requires upfront expert curation investment; this cost is not reflected in the benchmark's zero-build-cost assumption.
- Ground truth derived from DAG edges may not capture all valid natural language answers.
- All systems use the same LLM (Claude Sonnet 4.6); results may differ with other models.

9 Conclusion

We presented the CKG Benchmark, the first large-scale comparison of RAG, GraphRAG, and Compressed Knowledge Graphs across 25 educational domains. Our key findings are:

1. **RDS and Hop-Depth F1** are practical additions to IR evaluation that jointly measure quality and efficiency—metrics missing from existing benchmarks like BEIR and RAGAS.
2. **CKG outperforms RAG and GraphRAG** on structured domain retrieval, particularly on multi-hop dependency queries where explicit edges prevent the transitive inference failures inherent in embedding-based approaches.
3. **Pre-structured knowledge beats dynamically extracted graphs** when the domain is stable and expert-curated taxonomies are available.
4. **The McCreary Corpus** is a reproducible, citable multi-domain benchmark with standardized schema across 25 domains.

Future Work. We identify three directions for future research:

- Extending the benchmark to non-educational domains (legal, medical, financial) to test generalizability.
- Hybrid architectures that combine CKG structure with RAG's ability to retrieve rich contextual information.
- Automated CKG construction from unstructured text, which would eliminate the expert curation requirement.

Open Benchmark. The complete benchmark—corpus, queries, evaluation harness, and results—is released at `https://huggingface.co/datasets/graphify-md/ckg-benchmark` under CC BY 4.0 (data) and MIT (code). We invite the community to add domains, systems, and metrics.

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